ROGERS MUKIIBI

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Personal Profile

Electrical Engineering graduate from Makerere University, with applied experience across power systems, machine learning systems, and IoT. Co-authored an IEEE Access paper on an IoT-enabled multimodal honeybee monitoring system. Built machine learning models for solar irradiance estimation in Uganda, and a RAG-based chatbot at Xeno Investments. Hardware and simulation work runs from PCB design to distribution network modelling in DIgSILENT PowerFactory. Works in Python, C++, and SQL, and applies QUBO and QAOA formulations to distribution network control problems.

Education

Makerere University — **Kampala, Uganda**
BSc Electrical Engineering — *January 2022 – February 2026*

WorldQuant University — **Online**
Data Science Certificate — *December 2023 – January 2024*

Crested Secondary School — **Kampala, Uganda**
UACE - Physics, Chemistry, Mathematics — *February 2019 – April 2021*

Experience

Associate R&D Power Systems Engineer
Auri Energy — *July 2025 – present*

- Developed and validated power system models of distribution networks under varying load and fault conditions.
- Analysed voltage sags and swells, harmonic distortion up to the 40th order, and voltage unbalance against IEC standards.
- Contributed to the design of SCADA-based monitoring systems for real-time grid visibility.
- Supported digital twin development of distribution networks, combining simulation results with operational data.

AI Software Development Intern
Xeno Investments — *July 2025 – August 2025*

- Built a RAG-based chatbot for customer queries on Xeno Investments services, with ChromaDB vector search and LLM response generation.
- Added intent classification to keep simple queries away from the LLM, and session memory with latency logging.
- Designed an embedding pipeline that runs Google *gemini-embedding-001* or local *BAAI/bge-m3* models.

Research Assistant
Marconi Machine Learning Laboratory — *April 2024 – August 2025*

- Developed and deployed AI models that estimate solar irradiation for Uganda, to address the 3% to 20% positive bias of satellite data (CAMS RAD, NASA Power).
- Worked with GIZ and the Ministry of Energy to develop a free, open-source solar irradiation dataset fine-tuned for Uganda.

Undergraduate Intern — Portfolio
Internet of Things - Research & Applications Laboratory (IoT-RA) — *November 2023 – March 2024*

- Designed and fabricated a **2-layer PCB** for a smart beehive monitoring system, which cut assembly time by 83%, from 2 hours to 20 minutes.
- Integrated an ADXL345 accelerometer with a Raspberry Pi and built an **FFT** pipeline that samples hive vibrations at 10 kHz in real time.
- Co-authored the resulting **IEEE Access** paper, listed under Publications.

Undergraduate Intern
netLabs!UG — *June 2023 – August 2023*

- Designed and built **Proxmox-based server clusters** from electronic waste, extending hardware lifespan.
- Deployed **Docker containerised applications**, such as Blender, to give students without capable computers cloud-based access to design software.

Publications

2026 A. Turyagyenda, A. Katumba, R. Akol, M. Nsabagwa, M. E. Mkiramweni, and **R. Mukiibi**, *"A Smart IoT-enabled Multimodal System for Honeybee Keepers in Uganda,"* **IEEE Access**, doi: 10.1109/ACCESS.2026.3709045 (Early Access).

Projects

QubitBoard: Browser-Based Quantum Circuit Simulator

GitHub: rogersmukiibi/QubitBoard *August 2026 – present*

- Forked and maintain QubitBoard, a drag-and-drop simulator for circuits of up to 16 qubits, built on Craig Gidney's open-source Quirk.
- Built a code panel in TypeScript that converts a circuit to Cirq, OpenQASM 3, or a custom circuit language, and back, with a syntax-highlighting editor.

Skills

Power Systems: DIgSILENT PowerFactory, MATLAB/Simulink, load flow, voltage profiles, harmonic analysis up to the 40th order, voltage unbalance, SCADA integration, power quality, distribution systems.
Quantum Computing: Qiskit, Cirq, QUBO formulation, QAOA, circuit execution on IBM Quantum hardware (ibm_fez, 156-qubit Heron r2).
Software & Simulation: Python, Rust, C/C++, numerical simulation, digital twins, FFT, time-series analysis, domain-specific chatbots.
Embedded & Hardware: PCB design, KiCad, Altium, LTSpice, Multisim, Quartus (AHDL), Lattice Radiant (VHDL), Arduino, Raspberry Pi, LoRa (Pycom), sensor interfacing, data acquisition.
Engineering Tools: AutoCAD Electrical, technical documentation, system specification.

Professional Memberships

IEEE Member, with the IEEE Power & Energy Society and the Industrial Electronics Society.

Volunteering

2024	**Project Proposal Reviewer**	*EPICS in IEEE*
2024	**Webmaster AEE Makerere Chapter**	*Association of Energy Engineers*
2023	**Student Ambassador IEEE IES Region 8**	*IEEE Industrial Electronics Society*
2022	**Secretary IEEE CAS Makerere Chapter**	*IEEE Circuits and Systems Young Professionals*

Awards

2026	**Qsolve Hackathon, Nairobi, Kenya**	First Runners Up, Finance Track
2026	**IEEE Region 8 Student Paper Contest**	Third Place, from 5 finalists of 40
2025	**DataCamp Data Scientist Certification**	DS0026002460117
2024	**Makerere Engineering Society Open Day**	First Runners Up
2022	**IEEE Xtreme Coding Competition**	Ranked 1st in Uganda